



MICROTECH Microplate Washer

Instruction Manual

Catalog Numbers
170-7164 (115 Vac)
170-7165 (230 Vac)

BIO-RAD

For Technical Service Call Your Local Bio-Rad Office or in the U.S. Call 1-800-4BIORAD (1-800-424-6723)

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GENERAL INFORMATION

Please read carefully before using the MICROTECH Microplate Strip Washer

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About this manual

This manual has been written for the laboratory technician and describes how to operate the plate washer.

This manual was designed to provide you with the necessary information to:

- Install the MICROTECH
- Operate the MICROTECH to suit your needs
- Perform basic maintenance procedures, including disinfection.

This manual will also describe all the features and specifications of the hardware and configuration software.

Important Notice

Before installing this unit, please read carefully the installation instructions. This instrument can only be used to wash 96-well microplates with 8 or 12 channel manifolds. The MICROTECH is intended for clinical diagnostic and research laboratory use and must be operated only by specialized personnel aware of the potential risks connected to the chemical and bacteriological agents normally used with this unit.

The Aerosol Protection Cover must always be closed when the instrument is operated to reduce the risk of contamination from ambient air. Since the unit is not completely sealed, this risk is not completely eliminated, and the operator should dress with appropriate safety protection when using hazardous substances or biological agents.

While the unit is connected to the power, it is dangerous to operate without the protective cover. Operation without the cover may be performed only by specialized and trained service personnel. Contact your local Bio-Rad representative if service is required.

SAFETY INFORMATION

Transformer

Warning: Verify the correct voltage of the transformer before instrument operation. The primary voltage of the transformer is indicated on a label above the power cord socket (230 VAC or 115 VAC).

Serious damage to the instrument may occur if fluid penetrates the instrument protective case. The electrical supply **MUST NOT** be reconnected until the instrument has been fully inspected by a qualified technician or service engineer.

Warning : Spillages

In some operating modes (e.g. performing dispense mode or purge mode) it is possible to overfill the wells of a microplate. This can result in spillage of liquids and/or contamination with hazardous substances. Care should be taken to ensure that overfilling does not occur.

If environmentally hazardous fluid is spilled, the instrument must first be isolated from the electrical supply and physically isolated from the working environment by placing it in a fume hood or similar location. Contact the nearest service center for assistance and advise them of the health hazards associated with the spilled materials before servicing the equipment.

Warning : Incorrect Operation

Operating this equipment in ways other than detailed in this manual may impair the protection provided by the instrument.

Warning : Do not operate with the top cover open.
Do not poke fingers, lab supplies or other implements into the washer.
Leave the area in front of the washer free to avoid jamming of the plate carrier.
Wait until the plate carrier has stopped moving before removing or installing the microplate.

ENSURE that all connections are tight and all packaging elements (foams,...) have been removed before operating the washer.

ENSURE that the main power cable is correctly wired. Color codes are as follows

Europe

Brown	Live (+)
Blue	Neutral (-)
Green/Yellow	Earth (Ground)

United States

Black	Live (+)
White	Neutral (-)
Green	Earth (Ground)

Safety Symbols

The following alarm and information symbols can be found in several places in the washer.

Please read this manual carefully before using this instrument.

 Direct Current

 Alternating current

 Earth (Ground) Terminal

 Protective Earth (Ground) Terminal

 On

 Off

SECTION 1 INSTALLATION

1.1 GENERAL

Each instrument has been carefully tested before shipment. When opening the instrument, immediately verify that packaging is in good condition. Contact Bio-Rad Laboratories immediately to file a claim for any damages.

1.2 UNPACKING

Packing list

Please refer to the separate packing list provided with your instrument.

The package should contain the following:

- 1 One washer.
- 2 One external transformer.
- 3 One maintenance kit.
- 4 One power cord.
- 5 One user manual.
- 6 One Warranty Card
- 7 One Wash bottle (2 liter).
- 8 One Rinse bottle (2 liter).
- 9 One Waste bottle (2 liter).
- 10 One set of external tubes and tube connectors.
- 11 One hydrophobic filter.
- 12 Three separate bottle racks.

Unpacking instruction

Please refer to the separate INSTALLATION instruction enclosed with your washer (one page).

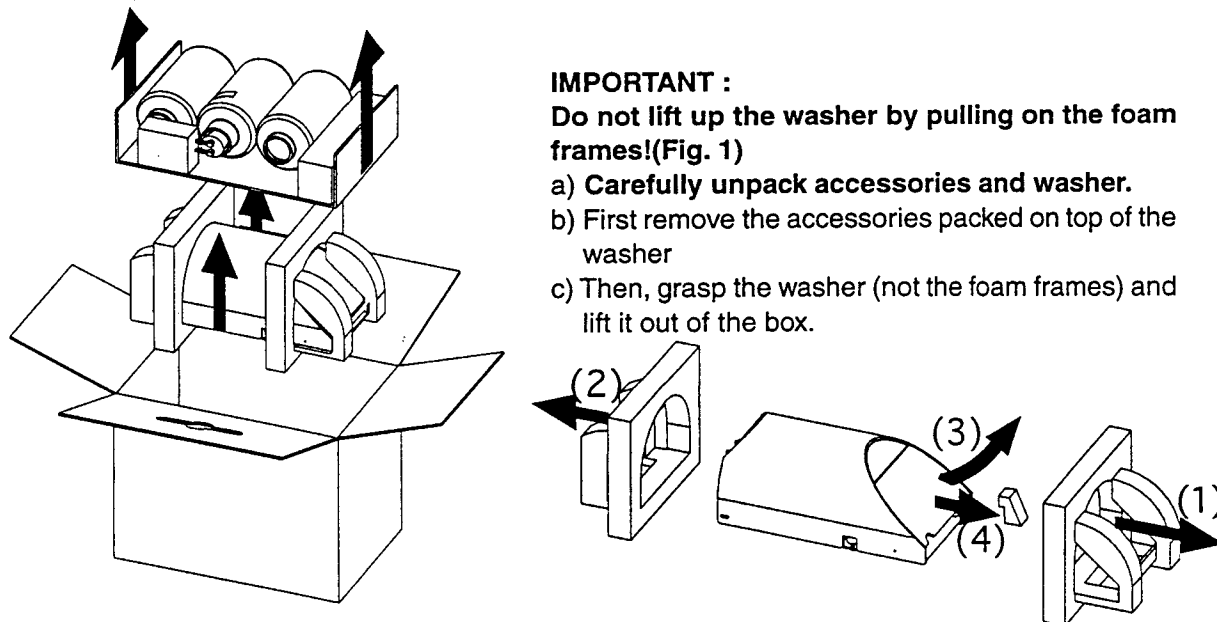


Fig. 1 Unpacking the washer.

- d) Remove the foam frames (1) and (2), remove plastic bag and place the washer on stable, horizontal surface.
- e) In washer working area:
 - Lift up aerosol cover (3).
 - Remove the foam protector (4) blocking the manifold.

1.3 INSTALLATION AND CONNECTION

Refer to the instructions enclosed in the fluidics kit for tube connection.

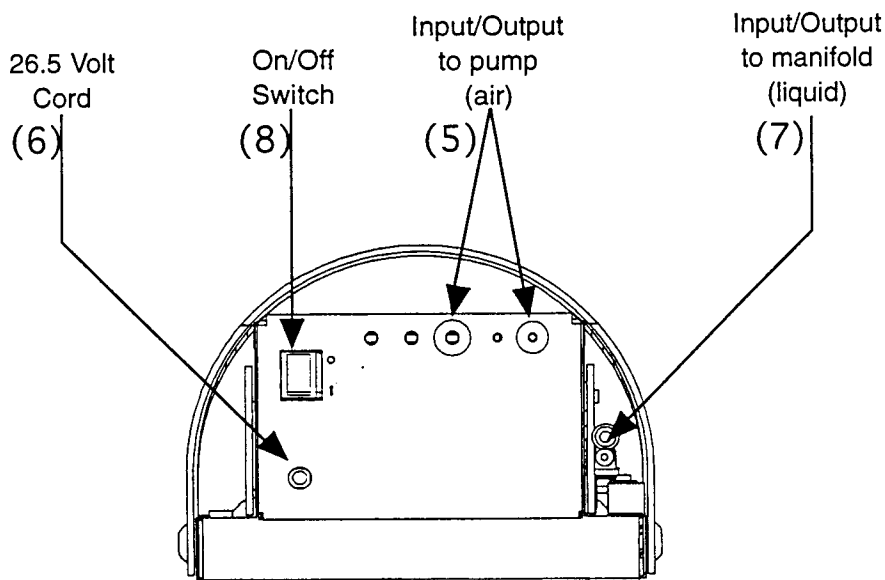


Fig. 2 Rear view

a) All tubes and connections are conveniently color coded for easy assembly.

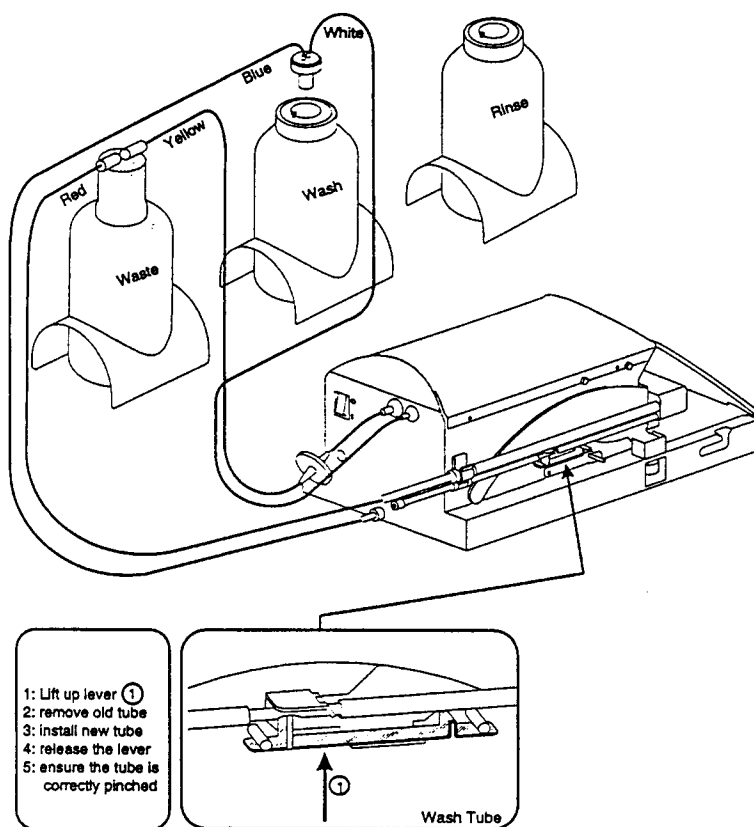


Fig. 3 Fluidic connection of the washer and blue manifold tube replacement procedure

1) Waste bottle connection :

- a) Unroll the red tube from the washer's left side (7) and connect it to red connector on Waste bottle (Fig. 2).
- b) Connect the 2 yellow tubes to both ends of the hydrophobic filter (supplied with the fluidics kit) and place this connection between the WASTE bottle and washer (match colors).

2) Wash bottle connection :

- a) Connect the blue wash tube (supplied in the fluidics kit) to the blue tube on the left side of the washer. Connect this tube to the wash bottle with the blue dot.
- b) Connect the white tube from the rear instrument panel of the washer to the wash bottle (white dot). (Fig. 3).

3) Transformer connection

Prior to plugging the transformer into the mains socket :

- Verify that transformer's primary voltage complies with the mains.
- Connect jack plug to the washer (6) (on rear panel below On/Off Switch). (Fig. 2).
- Connect power cord to the transformer (Fig. 4), then to the mains wall socket.

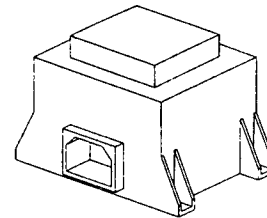


Fig. 4 Transformer

Primary : 110 VAC
or 230 VAC

**Verify voltage before
plugging to the mains!**

4) Powering up the Unit

- Turn the washer ON (On/Off switch on rear panel (8). (Fig. 2) While the plate carrier and manifold are moving to their home position, an autotest is performed (all the diodes are on orange as well as the LED display). (Fig. 5) After 1 second, the keyboard is displayed as follows (Fig. 6):

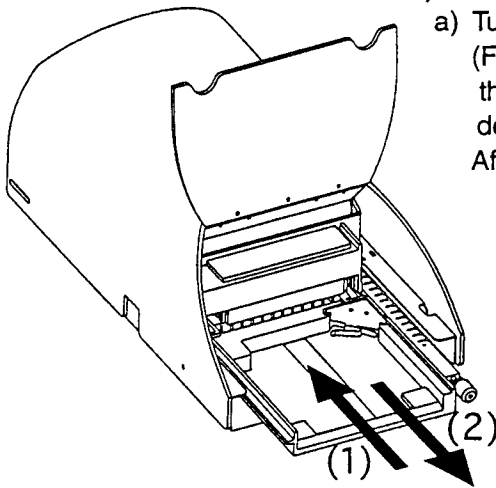


Fig. 5 Plate carrier motion at turning on.

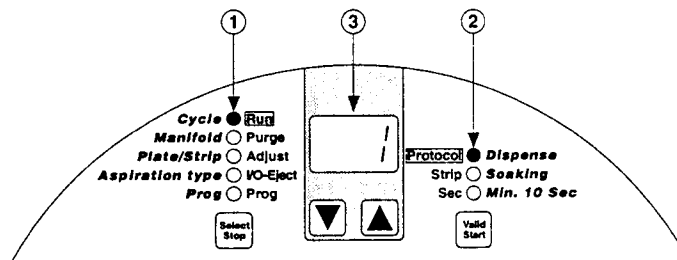


Fig. 6 Display at first turning on.

5) Priming the system

- Fill the Rinse bottle with distilled water (diH₂O) and prime the hydraulic circuit with at least 5 RINSE operations (see Purge: rinsing & priming in section 3.6). The MICROTECH washer is ready to use.

SECTION 2 SPECIFICATIONS

2.1 GENERAL VIEW

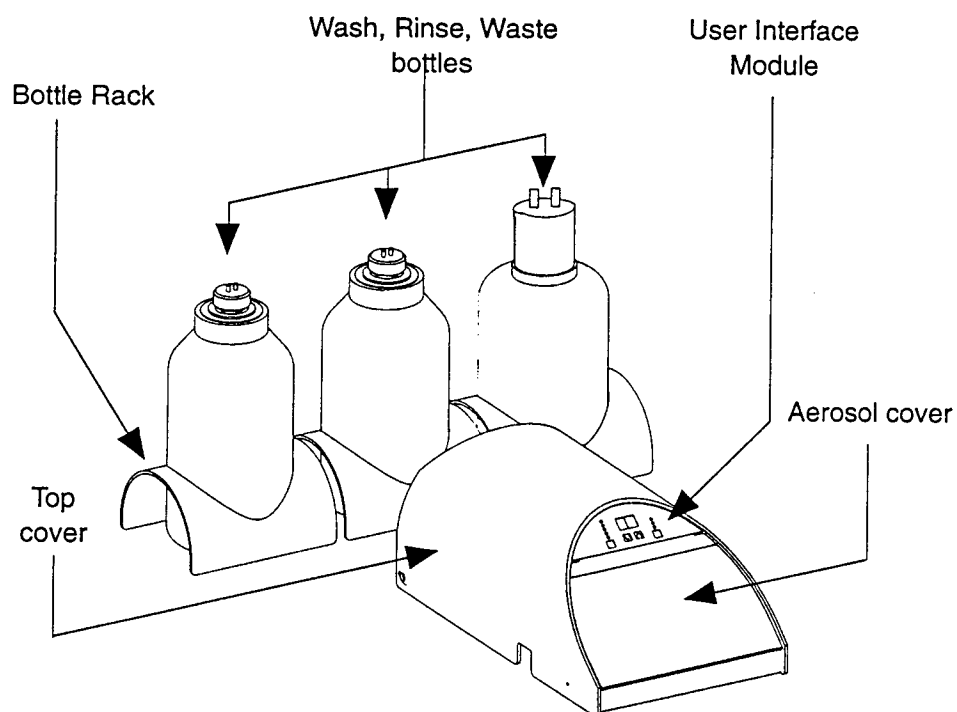


Fig. 7 General view

Fluidic circuit

Components :

- a) One bottle (2 liter) for the Wash solution.
- b) One bottle (2 liter) for the Rinse solution.
- c) One bottle (2 liter) collecting the Waste liquids.
- d) One integrated pump providing pressure for dispensing circuit (Wash/Rinse bottles) and vacuum for aspiration circuit (Waste bottle).

- e) All tubes between the bottles and washer (also inside washer) are color coded for easy recognition.
- f) Quick coupling connection on Waste bottle cap for safe handling of the waste bottle.

Electronic and Software

A microcontroller on a CPU board controls the Wash protocols and the other sequences as well as the communications with the User Interface Module (Keyboard).

The user friendly software features the following functions :

RUN Mode

Select and Run a Wash protocol.

PURGE Mode

Rinses the fluidic circuit with Rinse solution or Primes the fluidic circuit with Wash solution.

ADJUST Mode

Allows the user to manually adjust the overflow position and the horizontal position of the aspirating needle.

In/Out-Eject Mode

Moves the Plate carrier In/ Out, or up to the Eject position to be removed from the washer.

PROG Mode

Provides access and possible modification of the main wash parameters of the 25 stored protocols.

SECONDARY PARAMETER Mode

Allows modification of the secondary parameters of the wash protocols.

MACHINE PARAMETER Mode

Modifies the instrument parameters.

QUICK TEST Mode

Operation of run Protocol 0 will check all the modules of the washer.

MAINTENANCE Mode

Will separately check all the different modules of the washer.

ERROR LIST

The instrument will detect possible errors. The buzzer will beep intermittently and the error number will be displayed.

2.2 SPECIFICATIONS

Electrical data

Supply voltage :	26.5 VAC
Frequency :	50 or 60 Hz
Consumption :	22 VA max
Fuses :	1.25 A Slow Blow
Transformer :	230 VAC or 115 VAC

Physical data

Width :	225 mm
Length :	380 mm
Height :	165 mm
Weight (washer alone) :	5 kg
Gross weight :	9.0 Kg
Operating conditions :	15-30 °C / 15-85 % R.H.

Hardware specifications

Manifold :	8 or 12 channel
Vacuum type :	membrane pump
Operating Pressure in Wash/Rinse bottle :	0.38 ± 0.03 Bar
Flow rate at dispensing needle :	400 µl/sec ± 5%
Volume of Waste bottle :	2 liter

2.3 SOFTWARE WASH PARAMETERS

Note : This section describes the wash protocol, method, and software parameters. Please read it before attemptation to program wash protocols.

Wash Protocol Definition

The Wash protocol (or kit) is built using one method made of two elementary sequences : **n** (number) of cycles of **Wash** followed by one **Aspiration**.

nW+A



A **Wash** sequence (**W**) is made up of one Aspiration immediately followed by one dispensing sequence during which the dispensing needles are filling the wells.



An **Aspiration** sequence (**A**) causes the aspirating needle to move downward until touching the well bottom. Then the aspirating needles move upward to a home position.

6 programmable parameters have to be set for each Wash protocol. These parameters are written in red on the Keyboard interface.

- 1) Manifold type.
- 2) Wash mode.
- 3) Aspiration type.
- 4) Dispensing time
- 5) Soak time.
- 6) Cycles number.

25 Wash protocols are predefined but may be changed (refer to Sec. 4).

In addition, 2 positions (overflow position and the horizontal position of the aspirating needle relative to the well) have to be adjusted manually for each wash protocol. See section 4 PROGRAMMING.

Programmable parameters

Manifold

8-channel



or

12-channel



Plate/Strip

Strip mode

or

Plate mode

Aspiration Type



STRIP mode :
The total method is applied on each strip prior to processing the next strip.



PLATE mode :
The elementary cycle is applied successively on all strips prior to applying next elementary cycle.

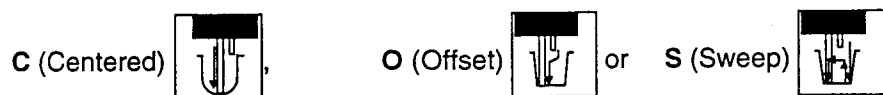
Example :
Protocol
3W +A

STRIP Mode

3W + A on 1st strip
3W + A on 2nd strip
etc...
3W + A on last strip

PLATE Mode

W on the whole plate
W on the whole plate
W on the whole plate
A on the whole plate



Depending on the microplate being processed, the needle can be shifted close to the wall of the well leaving to reduce residual volume.

The needle must remain centered (C) for U or V shaped wells and you may choose Centered (C), Offset (O) or Sweep (S) (Crosswise Aspiration) for flat bottomed wells.

Adjustment of centering or shifting of the aspirating needle in the well is performed manually using the horizontal adjusting knob and the Adjusting Mode (see details in section 4.3).

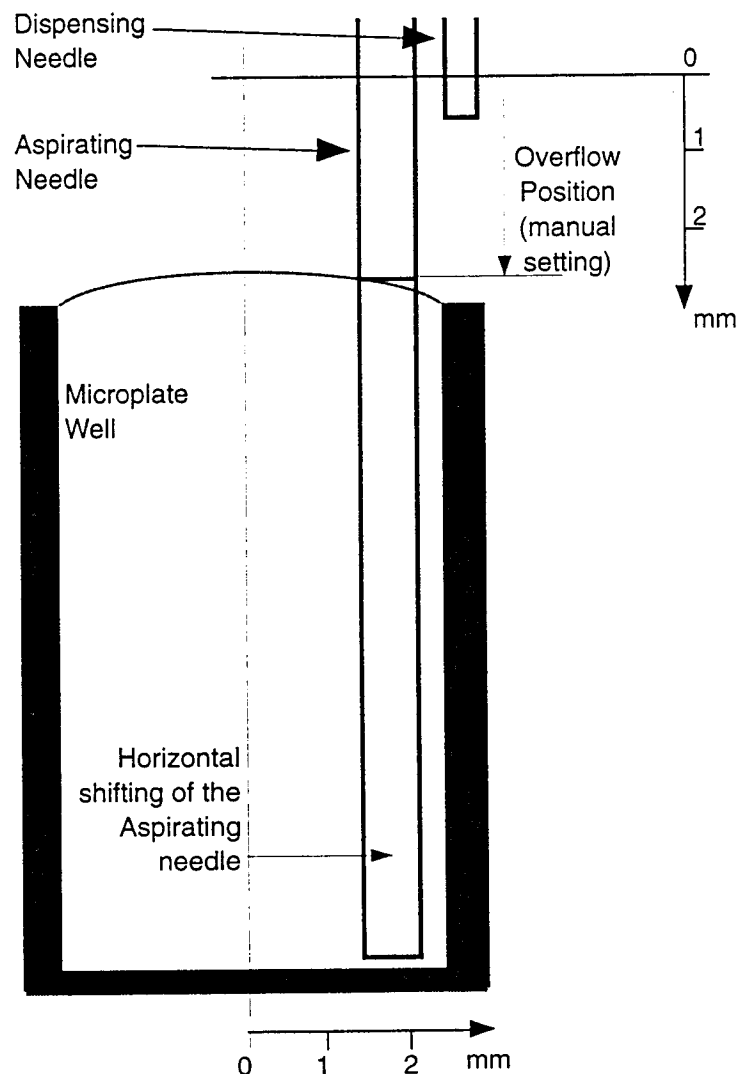


Fig. 8 Horizontal and Vertical positions to be set manually.

Dispense (time)

From 0 to 9.9 sec in 0.1 sec. increments.

Total time liquid is dispensed in the well by the dispensing needles.

The flow rate is 400 $\mu\text{l}/\text{sec.} \pm 5\%$ for one well.

Soaking (time)

In Strip mode

0 to 9.9 sec. in 0.1 sec. increments.

In Plate mode

0 to 9 mn 50 sec. in 10 sec.
increments

This is the elapsed time between two consecutive wash cycles for the same well. When the time necessary to process the whole plate (in Plate mode) is longer than the programmed Soak time, the next wash cycle will start immediately.

Cycle

0 to 9

nW + A

n refers to the number of consecutive wash cycles in a wash protocol.

Set the number of cycles to 0 to program a single Aspiration sequence.

SECTION 3 GETTING THE MICROTECH STARTED

3.1 USER INTERFACE MODULE DESCRIPTION

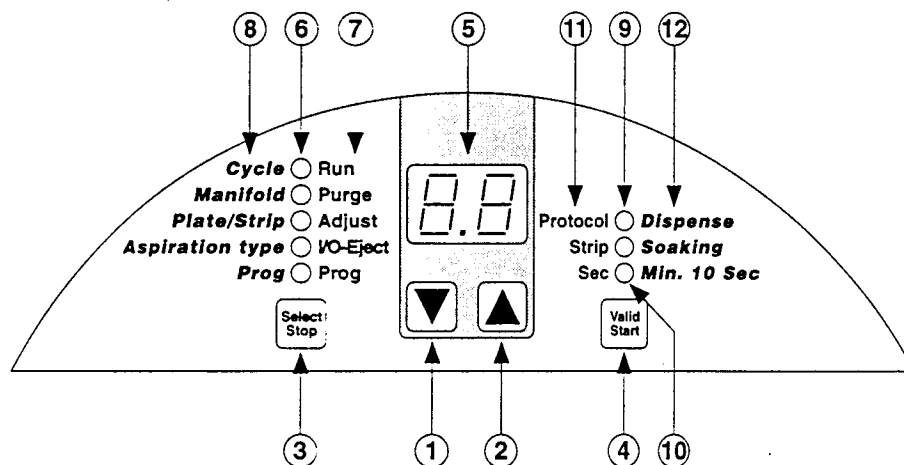


Fig. 9 User Interface Module

- 01 : ▼ key. Decrement of the displayed value.
- 02 : ▲ key. Increment of the displayed value.
- 03 : [Select/Stop] key. Select/Stop. Select a function or stop the on-going sequence.
- 04 : [Valid Start] key. Valid/Start. Validate the selection and/or Start the sequence.
- 05 : Double display, 7 LED segments. Display of protocol number, values of programmed parameters.
- 06 : 5 bicolored LED's (green in general mode). Display of function activation.
- 07 : Description of the general mode (printed in green).
- 08 : Description of the programming mode (printed in red).
- 09 : 2 bicolored LED's (green in general mode). Display of parameters used in Run mode.
- 10 : 1 bicolored LED (green in general mode). Display of time units used in programming mode.
- 11 : Description of the general mode (printed in green).
- 12 : Description of the programming mode (printed in red).

3.2 TURNING THE UNIT ON

In order to facilitate reading and understanding of this manual, the following symbols are used : ○ Led off, ● Led on, ⊕ flickering Led, □ for the corresponding active function.

Turn the unit on by means of the power switch on the rear panel.

Display (Fig. 10) is as follows :

- 1- Run, green Led on
- 2- Protocol, green Led on
- 3- Number of the last used protocol

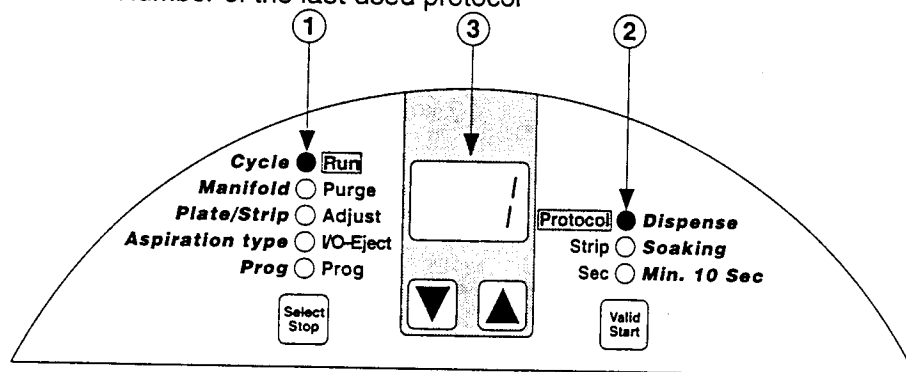


Fig. 10 Display when turning on.

Note : Prime the circuit with distilled water at least five (5) times (See section 3.6 Purge: Rinsing & Priming). Ensure the caps on the bottles are correctly connected and that there is no leakage.

3.3 LOADING THE MICROPLATE

Type of microplate :
any Flat, «V» or «U» shaped Microplate in a 96 - well format may be used.
The microplate frame must be rigid.

Eight (8) Mode configuration

Using the 8-channel manifold

Microplate
A1 Corner

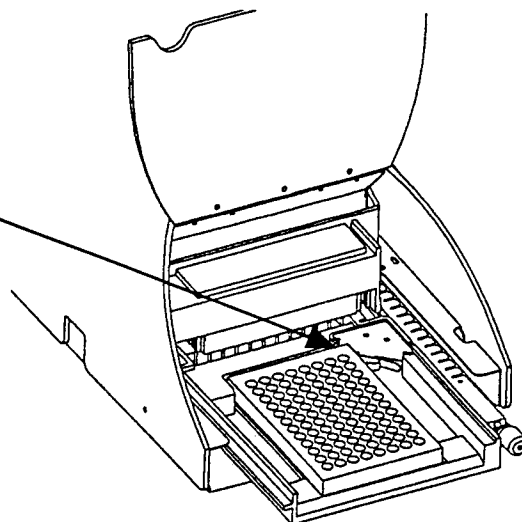


Fig. 11 8 Mode Configuration

Twelve (12) Mode configuration

Using the 12-channel manifold

Microplate
A12 Corner

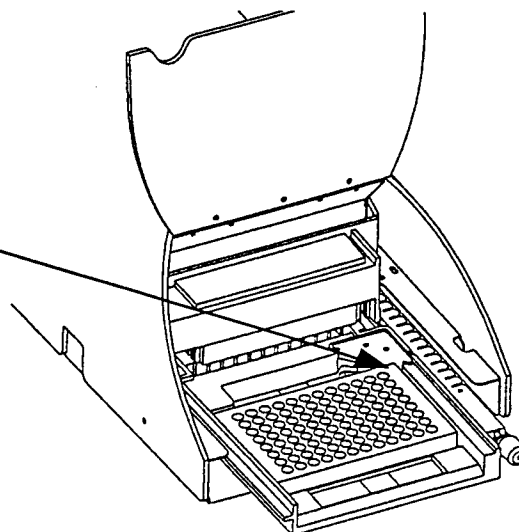


Fig. 12 12 Mode Configuration

3.4 RUN A WASH PROTOCOL

Select Wash protocol number (from 1 to 25) using ▼ and ▲ keys. Validate the selection by pressing the  Valid/Start key.

On the user interface module (Fig. 13):

- 1- Green LED **Run** is On,
- 2- Green LED **Strip** is On.
- 3- Strip number (indicates by default the strip number of the last protocol)

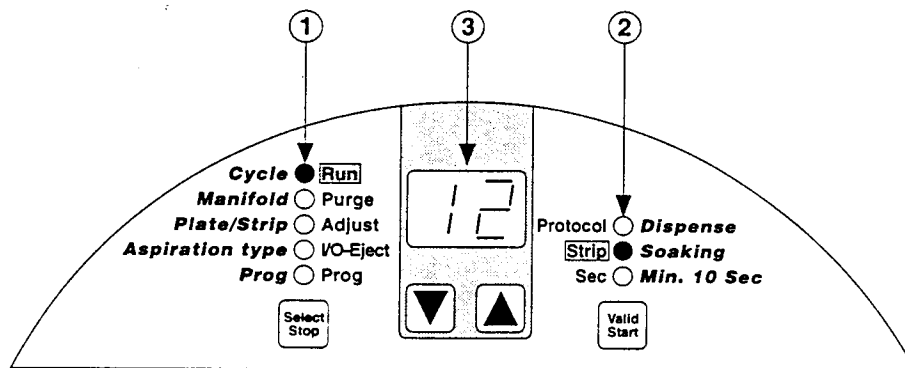


Fig. 13 Display of Strip number.

The above run sequence can be cancelled by pressing the  Select/Stop key at any time during a run. The washer will automatically return to the initial status. For 8-channel manifold protocols, select the number of washed strips between 1 & 12.

For 12-channel manifold protocols, select the number of washed strips between A & H.

Once the  Valid/Start key is pressed the display (Fig. 14) is as follows :

- 1- Green LED **Run** is flickering,
- 2- Green LED **Protocol** is On.
- 3- Protocol number is displayed.

The selected protocol will run to completion. When finished, the washer will return to the initial status.

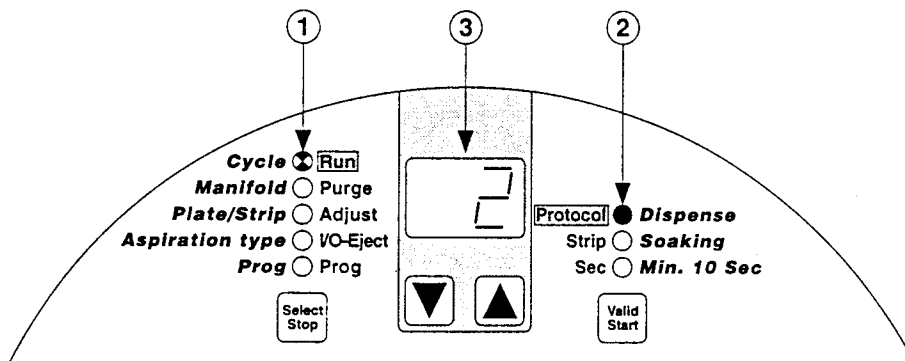


Fig. 14 Display with Wash protocol in progress

3.5 INTERRUPTION OF A WASH PROTOCOL

Press the  Select/Stop key to stop the wash protocol in progress.

Display (Fig. 15) is as follows :

- 1- Green LED **Run** is flickering,
- 2- Green LED **Protocol** is On.
- 3- Displayed protocol number is flickering.

To resume the protocol, press the  Valid/Start key and it will resume the wash sequence at the point where it was stopped.

To stop the protocol definitively, press a second time on the  Select/Stop key. The washer will return back to initial status.

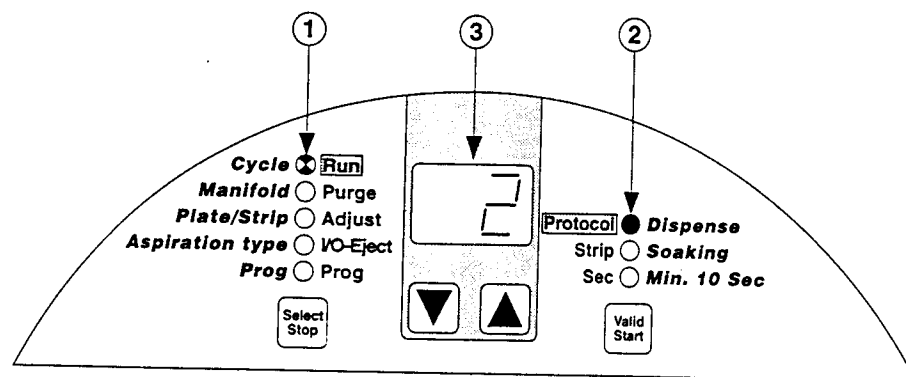


Fig. 15 Display with Wash protocol interrupted.

3.6 PURGE : RINSING & PRIMING

Important : The washer shall never be switched off with WASH solution inside tubing and manifold. Solid crystals from dried WASH solution can clog the dispensing needles and dispensing tube.

Rinsing

When the washer remains in standby for 10 minutes after the end of a wash protocol (or a priming), a display (Fig. 16) is shown prompting the user to perform a Rinse operation.

- 1- Green LED **Purge** is flickering,
- 2- Letter «r» for RINSE is displayed and flickering.

Remove the cap from the Wash bottle (remove slowly, bottle under pressure) and connect it on Rinse bottle; Start rinsing by pressing the  Valid/Start key. At the end of the rinse cycle, the washer returns to initial status. Reconnect the cap to the Wash bottle.

For Rinsing or Priming sequences, the plate carrier moves to Out position so as to place the rinsing tank below the manifold .

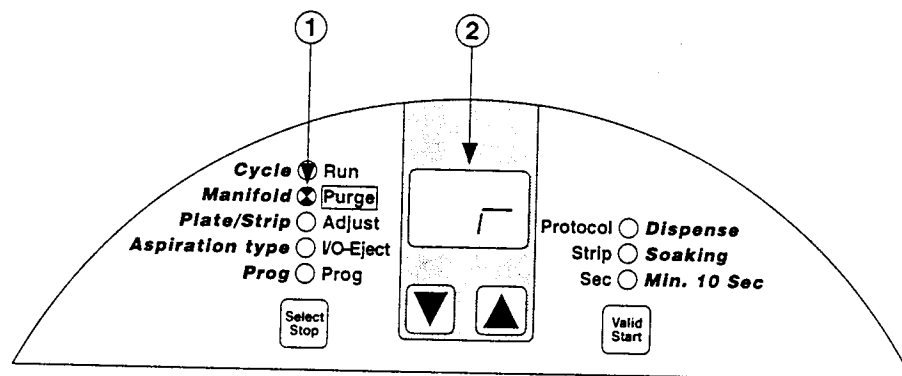


Fig. 16 Display of request for Rinsing.

Likewise, when turning the unit off after a Wash sequence, a beep will be emitted for few seconds to remind the user that a Rinsing operation must be performed.

Priming

Priming with Wash solution is possible. In Purge mode, Press the  or  keys to have the letter (P) displayed. Display (Fig. 17) is as follows :

When launching a wash protocol (Run mode) after a rinse operation, the washer will prime automatically with wash solution before starting to wash the plate.

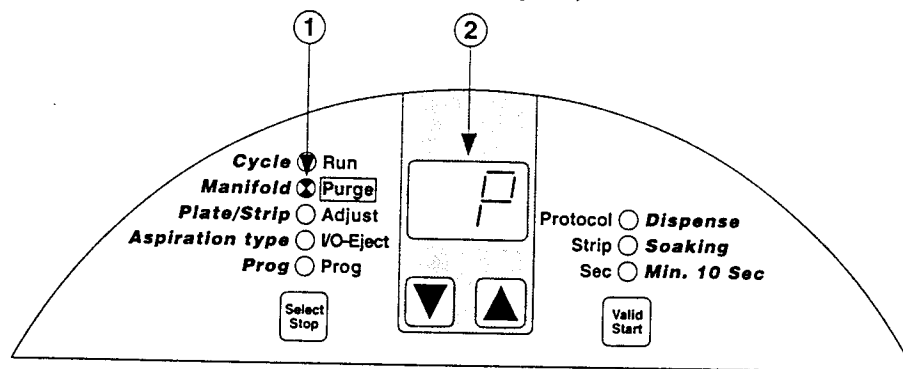


Fig. 17 Display of request for Priming

- 1- Green LED **Purge** is flickering,
- 2- Letter (**P**) for Prime is displayed and flickering,

Connect the washer to the Wash bottle and start priming by pressing the  Valid/Start key. At the end of the priming sequence, the washer returns to initial status.

3.7 IN/OUT-EJECT

This function allows the user to move the plate carrier In and Out and to remove the carrier for cleaning.

Select the I/O-Eject function by pressing the  Select/Stop key. The display (Fig. 18) is as follows :

- 1- Green LED I/O-Eject is On,
- 2- Letter (**I**) for In, (**O**) for Out or (**E**) for Eject is On.

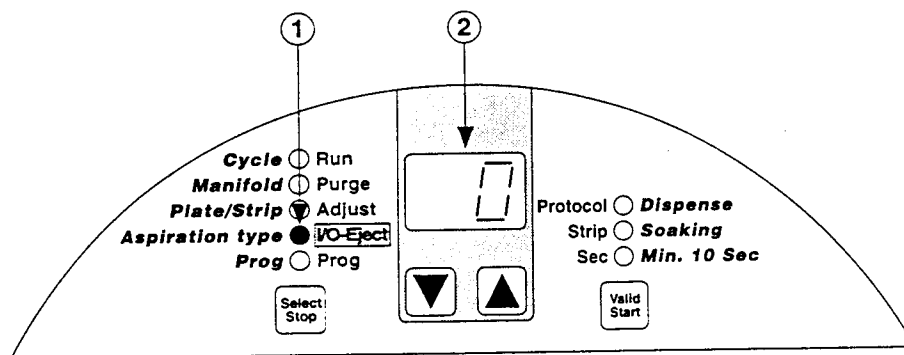


Fig. 18 Display of I/O-Eject mode.

Select the desired position with the  or  keys.

a) When selecting the In motion (**I**), pressing the  Valid/Start key causes the plate carrier to move inside the washer at rest position. The letter (**O**) is then displayed.

b) When selecting the Out motion (**O**), pressing the  Valid/Start key causes the plate carrier to move out and the letter (**I**) is displayed.

c) When selecting the Eject motion (**E**), pressing the  Valid/Start key causes the plate carrier to move out to Eject position.

The plate carrier can be easily removed from the washer.

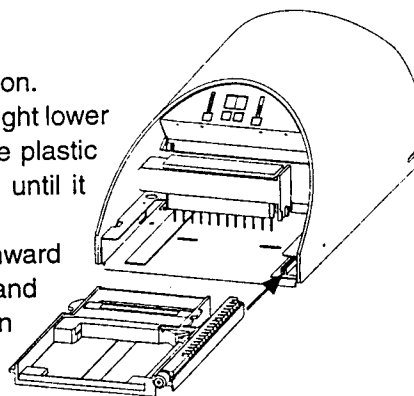
Letter (**I**) is then displayed.

To reinstall the plate carrier :

In the I/O-Eject mode, select the In (**I**) motion.

Place the plate carrier on the washer. The right lower edge of the carrier must be inserted in the plastic groove of the washer. Push the carrier in until it has stopped.

Press the  Valid/Start key to start the inward motion. Push the plate carrier with your hand to put it into gear. The plate will be moving In and Out to rest position. At the end of the motion the Letter (**I**) is displayed.



3.8 ADJUST MODE

This mode allows the user to set the overflow position and the horizontal position of the aspirating needle in the microplate well. This adjustment is done manually by means of two adjusting knobs.

The Adjust Mode is detailed in Programming section 4 (see section 4.3)

SECTION 4 PROGRAMMING

4.1 MAIN PARAMETERS

Select the Prog mode by pressing the  Select/Stop key.

The programming mode allows access to and possible modification of the main parameters of a wash protocol.

The display (Fig. 19) is as follows :

- 1- Green LED Prog is On,
- 2- Green LED Protocol is On,
- 3- Number of last protocol is displayed.

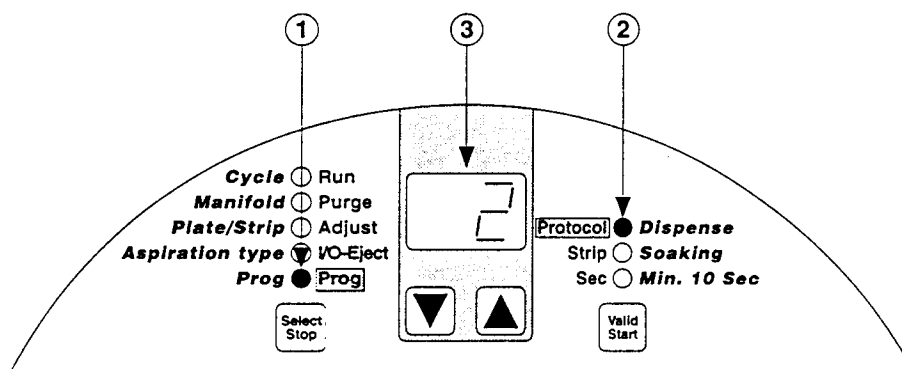


Fig. 19 Display at Prog mode selection

Press the  or  keys to display the desired wash protocol number. Confirm the selection by pressing the  Valid / Start key. The display (Fig. 20) is as follows :

- 1- Red LED Prog is flickering,
- 2- Red LED Manifold is on,
- 3- Manifold type is displayed (8).

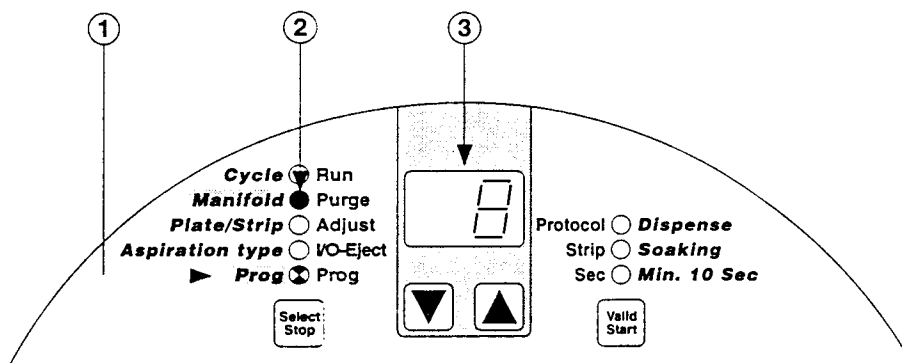


Fig. 20 Display of 1st parameter in Prog mode

Press the  Select/Stop key to access the different parameters of the protocol.

The wash protocol parameters are as follows :

- Manifold type : 8 or 12.
- Wash mode : (**P**) Plate or (**S**) Strip.
- Aspiration type : (**C**) Centered, (**O**) Offset, (**S**) Sweep (crosswise).
- Dispensing time : from 0 to 9.9 second.
- Soak time : depending on the wash mode :
0 to 9.9 second in strip mode.
0 to 9 minute and 50 second in plate mode.
- Cycles number : 0 to 9.

For every displayed parameter, the associated red LED is on.

For time parameters (2) (dispense time and soak time), a second LED for the time unit is on; its color (green or red) depends on the applicable time unit.

green LED : unit = second . decimal = tenth of second

Ex. : 3 . 5 = 3.5 seconds.

red LED : unit = minute . decimal = ten second increments.

Ex. : 3 . 5 = 3 minutes and 50 seconds.

For example, when displaying the dispense time, display (Fig. 21) is as follows :

- 1- Red LED Prog is flickering.
- 2- Red LED Dispense is on.
- 3- Dispense time is displayed (3.5 sec).

Press the  or  keys to modify the parameter value. When the desired value

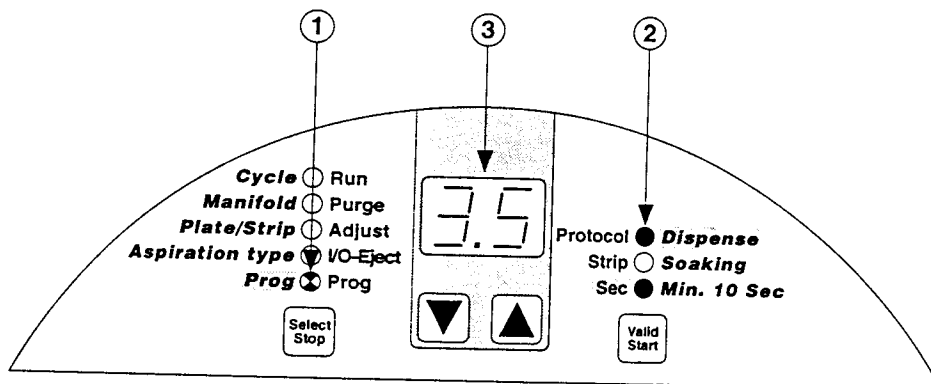


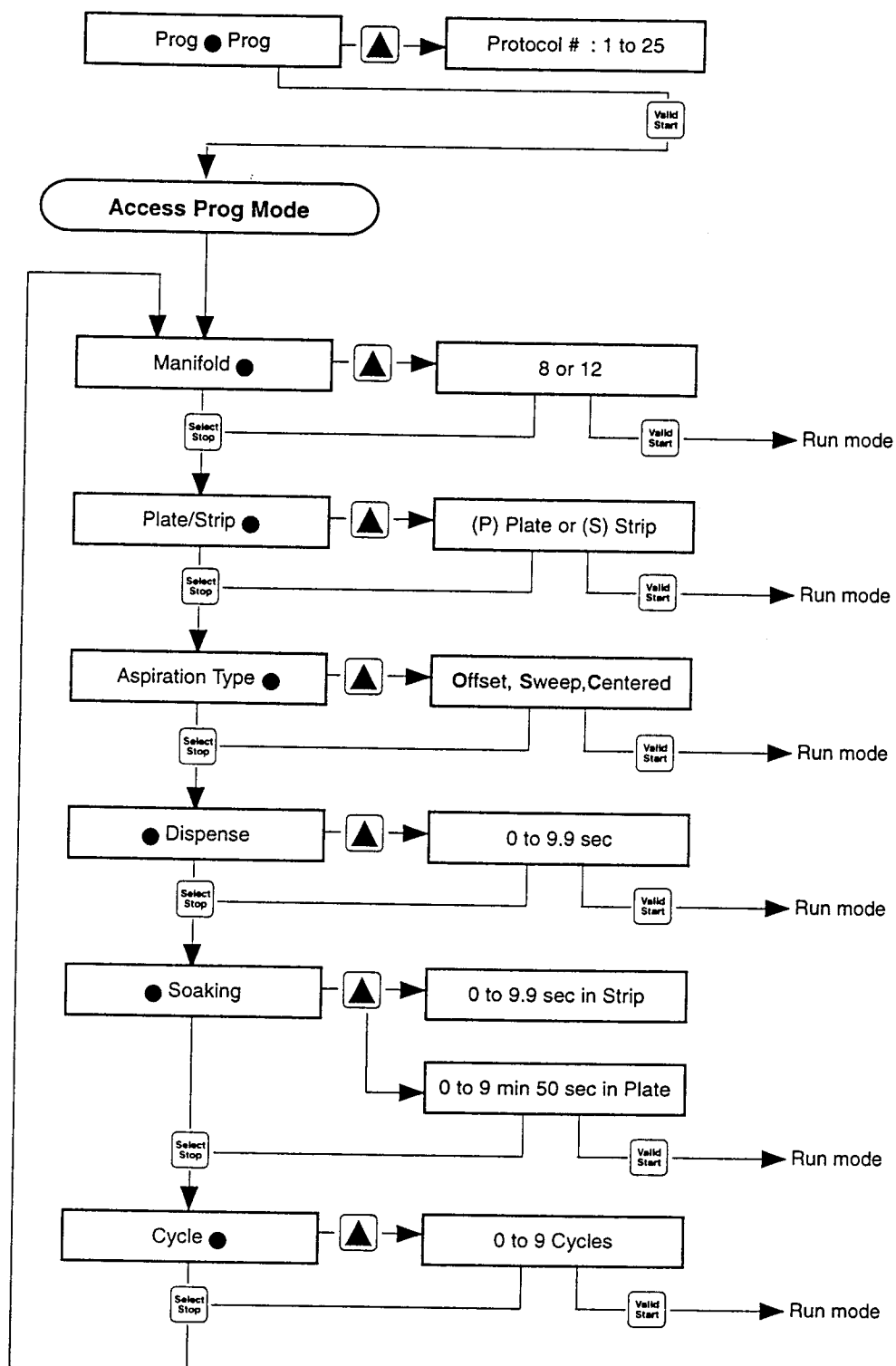
Fig. 21 Display of the dispense time (in second).

is displayed, it can be validated either by :

- Using the  Select/Stop key to cause the new value to be saved and select the next parameter.
- Using the  Valid/Start key to cause the new value to be saved and exit the programming mode. Display is then same in fig. 4-1.

Note : To escape from the programming mode at any time, press the  Valid/Start key.

4.2 PROGRAMMING FLOW CHART



4.3 ADJUST MODE

This mode allows the user to set the overflow position and the horizontal position of the aspirating needle in the microplate well. This adjustment is performed manually by two adjusting knobs. Push the cover forward to access the overflow position knob located on the left side.

The horizontal position knob is located on the front right corner of plate carrier. To modify the horizontal adjustment : push and turn the horizontal position knob. Select the Adjust mode by pressing the  Select/Stop key until (Fig. 22):

- 1- The green LED Adjust is On.

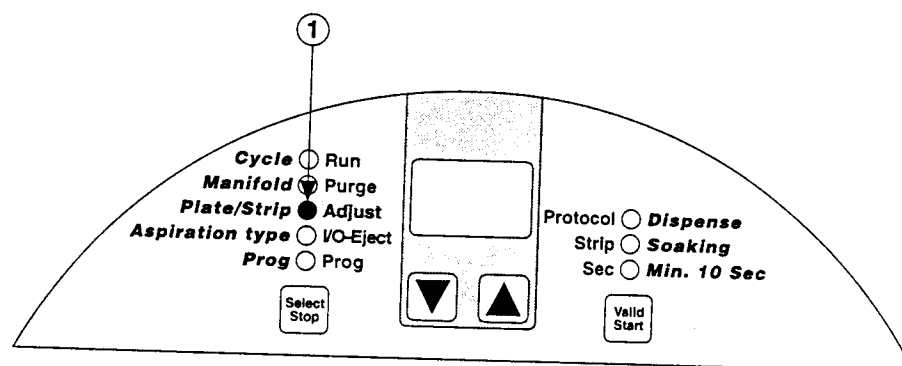


Fig. 22 Display at Adjust mode selection

Validate with the  Valid/Start key. Display (Fig. 23) is as follows :

- 1- Green LED Adjust is flickering,
- 2- Red LED Aspiration type is On,
- 3- Letter (O) for Offset is flickering.

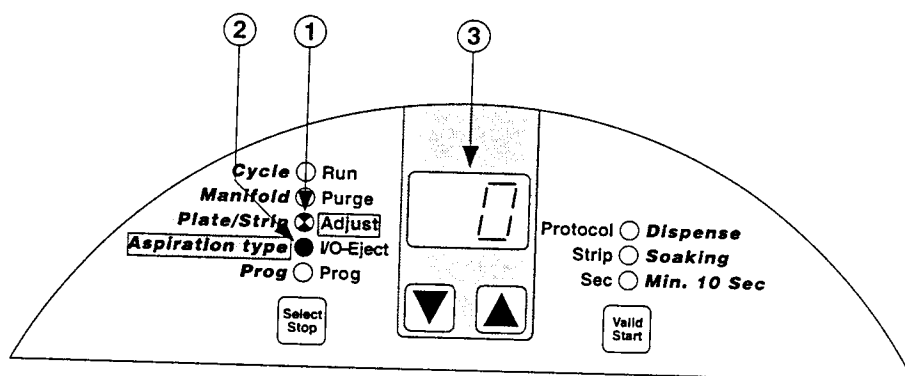


Fig. 23 Display of the Adjust mode with Aspiration type at Offset

The Offset letter (O) indicates that the aspirating needle is offset (or shifted) from the center of the well to optimize aspiration of residual volume in flat bottomed microplates.

The instrument positions the plate carrier to a compatible position for either 8- and 12-channel manifold and will move the manifold up to an overflow position.

At this point, the user can select the other two aspiration types that are available. Press the ∇ or $\blacktriangle$ keys to display the **(S)** Sweep or **(C)** for Center manifold positioning.

a) **(S)** Sweep: means sweep (crosswise aspiration).

The sweep aspiration utilizes the offset aspiration position by performing a second aspiration in the well on the opposite end from the original offset position. The manifold is lifted up to move from the first to the second aspiration position to avoid scratching the well bottom.

The sweep aspiration is recommended for achieving low residual volume with flat bottom wells.

b) **(C)** Center: means centered aspiration, to be used with U- or V-shaped wells. If the aspiration type is modified, confirm the modification by pressing the $\equiv$ Valid/Start key. This causes the plate carrier to be shifted the second horizontal position.

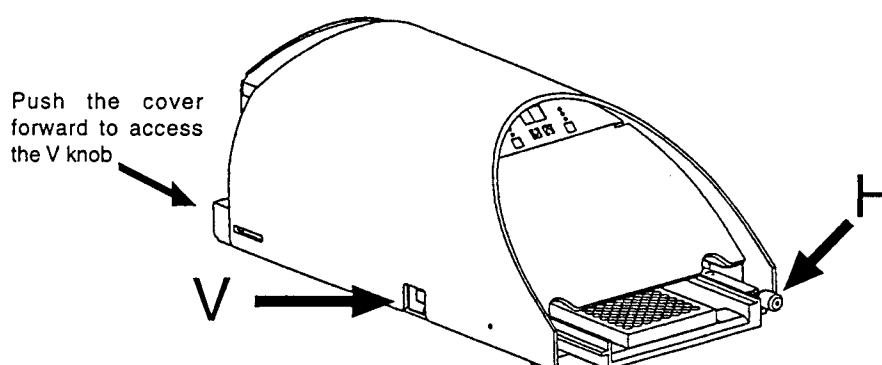


Fig. 24 The adjusting knobs on the washer

Vertical setting



Overflow
sequence

Adjust the overflow position by adjusting the vertical position (**V**) of the manifold using the overflow knob on the left side (Fig. 24).

Horizontal setting Aspiration



C

Adjust the horizontal position, (**H**) by adjusting the horizontal position knob located on the front right corner of the plate carrier (Fig. 24).

Select the aspiration key to select **(C)** Center, **(O)** Offset, **(S)** Sweep



O

a) With **(C)** Centered type: the aspirating needle must be positioned in the center of the well (Curved or V-bottom wells).

b) With **(O)** Offset type: the aspirating needle must be close to well's wall (Flat).

c) With **(S)** Sweep (or crosswise): the aspirating needle must be close to well's wall, like in Offset type (Flat).



S

Verify that the adjustment is correct by launching a down and up movement of the manifold by pressing the $\equiv$ Valid/Start key.

Note :

With Sweep **(S)** type (crosswise aspiration), press the $\equiv$ Valid/Start key a second time to move the plate carrier to the opposite end of the well.

Exit the ADJUST mode by pressing the $\equiv$ key. The plate carrier will move to its horizontal position and the display will return to mode selection.

4.4 ERROR LIST

Failure, mechanical jam, or a software problem may disrupt normal operations. An error number will flicker on the display while beeping intermittently. The table below indicates the error definition as well as a first level of trouble shooting.

To stop the error alarm, press the  Select/Stop key or the  Valid/Start key.

An error can appear at any time while the washer is in operation :

- a) While washing
 - i During an In/Out motion
 - ii During an Up/Down motion
- b) While rinsing
- c) While adjusting manifold/well positions
- d) While an In/Out motion of the plate carrier
- e) While programming or while in Maintenance mode
- f) While turning the washer on (Autotest)

	Description <i>When</i> <i>Trouble shooting</i>	Continuous beep with no display. with instrument powered up Check connection of flat wiring between the PCB display and CPU board. Turn the instrument on again and verify that all indicator lights (discrete LEDs and 7 segment LEDs) are turned on for a short time. Should there still be a continuous beep but light indicators turned on correctly, the failure is likely to be located on the CPU board or flat wiring. Please contact a Bio-Rad technical service agent for repair direction.
01	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	In/Out movement error at stroke end sensor Inward motion of plate carrier to home position (this motion is part of the initialisation motion). Home position (In) has not been detected by SH1I sensor within the expected time, whereas SH2R sensor has detected a rotation of the horizontal slotted disc. Contact a Bio-Rad technical service agent.
02	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	In/Out movement error at stroke end sensor At inside motion of plate carrier to home position (this motion is part of the initialisation motion) Home position (In) has not been detected by SH1I sensor within the expected time and SH2R sensor has not detected any rotation of the horizontal slotted disc. Contact a Bio-Rad technical service agent.
03	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	In/Out movement error during a slot to slot motion Outside motion of plate carrier to home position (this motion is part of the initialisation motion) During washing No signal found within the expected time at SH2R sensor (detecting the motion of the horizontal slotted disc) Contact a Bio-Rad technical service agent.

04	Description	Up/Down movement error during a motion to home position
	<i>When</i>	Up/Down movement initialisation During washing
	<i>Comments</i>	Home position has not been detected within the expected time at SV1I sensor (vertical home position sensor).
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
05	Description	Up/Down movement error
	<i>When</i>	During washing
	<i>Comments</i>	No signal found within the expected time at SV2R sensor (detecting the motion of the vertical perforated disc.
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
06	Description	Switching to 50 Hz for Up/Down motor impossible
	<i>When</i>	During autotest At every start of washing or rinsing
	<i>Comments</i>	Defective 50/60Hz switch relay (connecting an additional dephasing capacitor in parallel for use at 50Hz).
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
07	Description	Mains frequency too low.
	<i>When</i>	During autotest
	<i>Comments</i>	The frequency value measured is too low to ensure a correct running of the washer.
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
08	Description	Mains frequency too high.
	<i>When</i>	During autotest
	<i>Comments</i>	The frequency value measured is too high to ensure a correct running of the washer.
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
09	Description	Exceedingly high pressure on fluidic circuitry
	<i>When</i>	During washing During rinsing
	<i>Comments</i>	The high pressure threshold has been overrun on pressure sensor
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
10	Description	Exceedingly low pressure
	<i>When</i>	During washing During rinsing
	<i>Comments</i>	The low pressure threshold has not been detected by the pressure sensor within the expected time frame.
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.
11	Description	Defect on vacuum circuit
	<i>When</i>	During washing During rinsing
	<i>Comments</i>	Vacuum threshold has not been detected within the expected sensing time
	<i>Trouble shooting</i>	Contact a Bio-Rad technical service agent.

12	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	26.5V rectified supply voltage too high During autotest Measured value of the 26.5V rectified rectified above upper limit Contact a Bio-Rad technical service agent.
13	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	26.5V rectified supply voltage too low During autotest Measured value of the 26.5Vd rectified below lower limit Contact a Bio-Rad technical service agent.
14	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	26.5V regulated supply voltage too high During at pump startup Measured value of the 26.5V regulated rectified above upper limit Contact a Bio-Rad technical service agent.
15	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	26.5V regulated supply voltage too low During at pump startup Measured value of the 26.5V regulated below lower limit. Contact a Bio-Rad technical service agent.
16	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	EEPROM access or contents defect During Autotest The default value was not found with the microcontroller memory. Contact a Bio-Rad technical service agent.
17	Description <i>When</i> <i>Comments</i> <i>Trouble shooting</i>	EEPROM writing error Programming Parameter mode The writing of a new value in the microcontroller memory has failed. Contact a Bio-Rad technical service agent.
18	Description <i>When</i> <i>Trouble shooting</i>	Software error All modes Contact a Bio-Rad technical service agent.

SECTION 5 MAINTENANCE

5.1 CONSUMABLE AND SPARE PART LIST

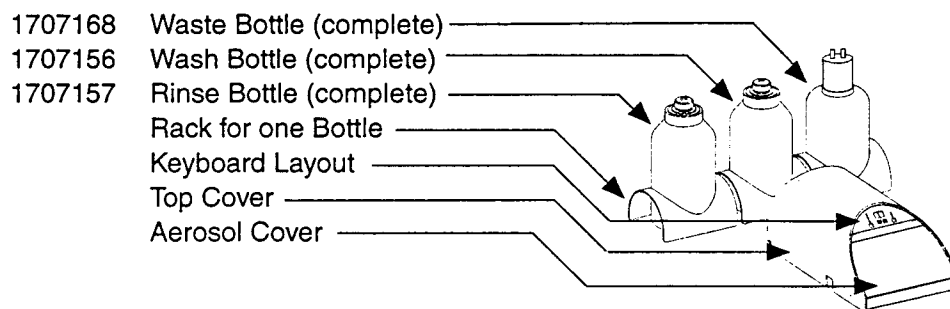


Fig. 25 General View

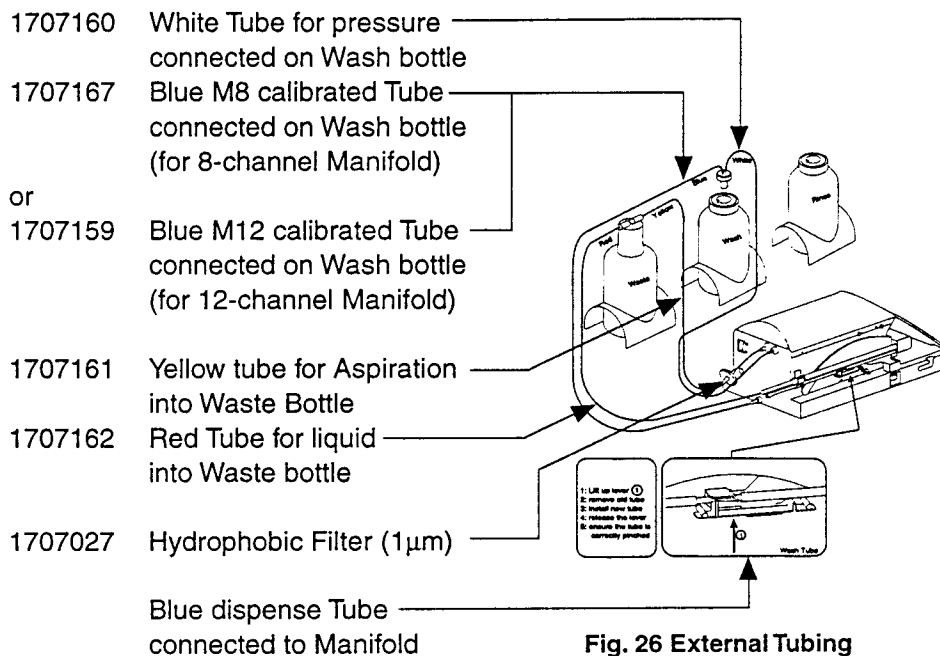


Fig. 26 External Tubing

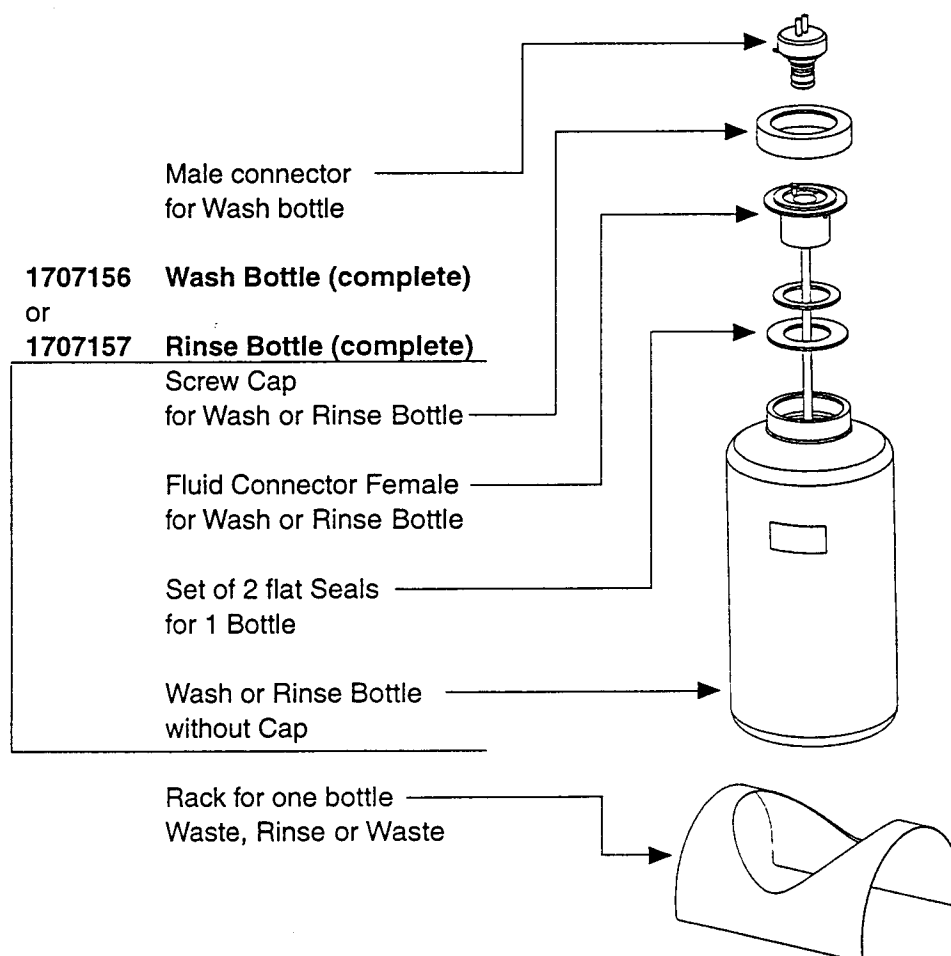


Fig. 27 Wash or Rinse bottle

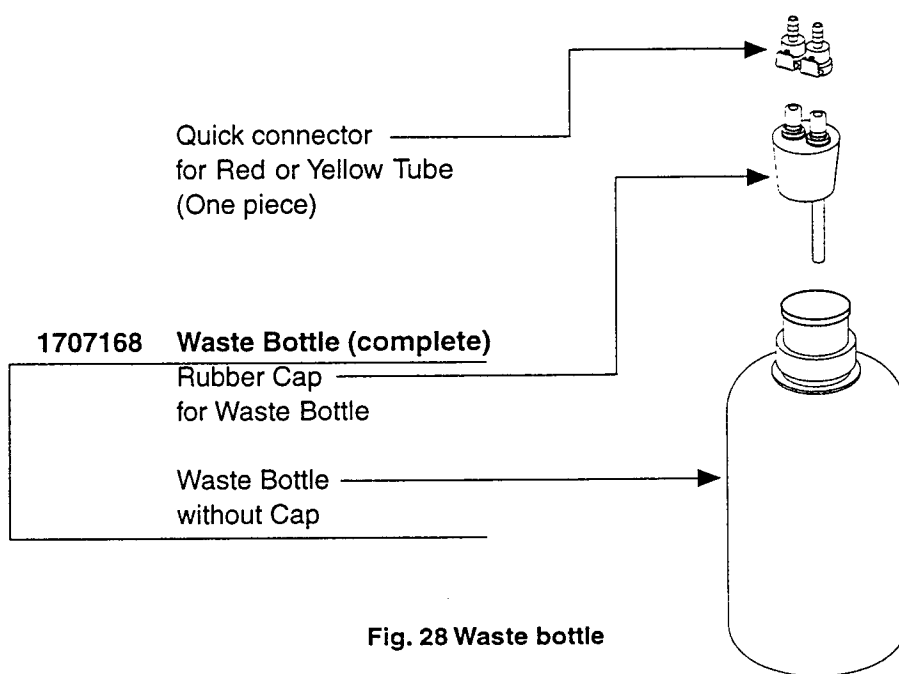


Fig. 28 Waste bottle

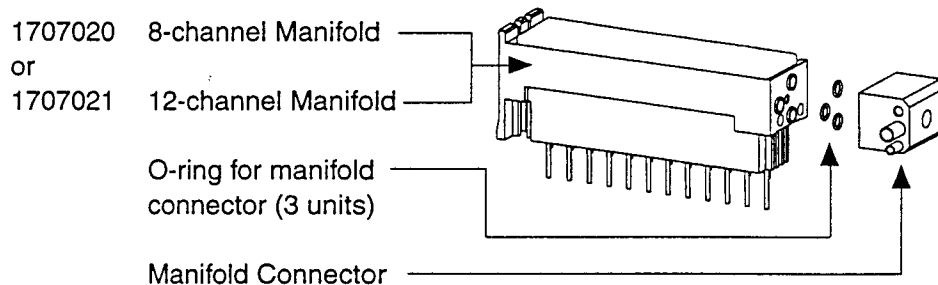


Fig. 29 Manifold (1)

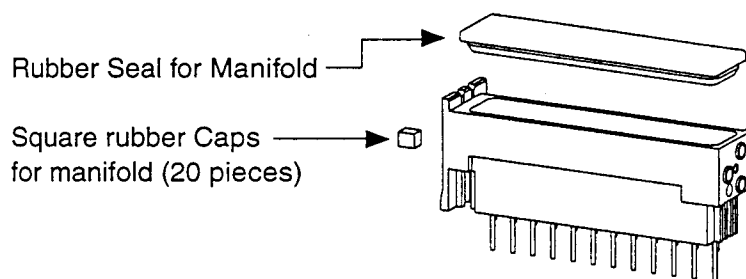


Fig. 30 Manifold (2)

1707154 Manifold Cleaning Kit

- (A) Silicone Cap to close dispensing chamber
- (B) Brush for cleaning dispensing Chamber
- (C) Small cleaning Pin for Dispensing Needles
- (D) Big cleaning Pin for Aspirating Needles
- (E) O-rings (3 pieces)

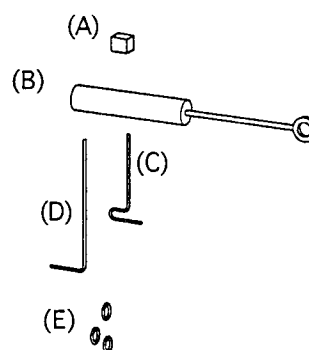


Fig. 31 Manifold Cleaning Kit

5.2 ROUTINE MAINTENANCE

Daily Maintenance

Prior to starting wash :

It is recommended to pour an appropriate disinfection solution in the bottom of the WASTE bottle prior connecting it to the washer. Thus, the infected liquid coming from the washer and going into the WASTE bottle will be deactivated.

IMPORTANT : Avoid bleach

Fill the rinse bottle with deionized water. Connect it to the washer. Open top cover and check that blue tube is not clogged in the area where pinched. Close top cover and launch a Rinse and check that manifold is not clogged.

Otherwise, change blue tube connected to manifold, clean manifold (Manifold cleaning).

When washing is terminated :

Rinse the washer. Use only deionized water for rinsing. Turn the washer off. Tubes and manifold will be kept wet with water that will prevent them from clogging.

Empty WASH bottle, rinse it thoroughly with distilled water.

Maintenance advice

Important : The MICROTECH washer should never be switched off with WASH solution inside tubing and manifold. Solid crystals from dried WASH solution will clog dispensing needles and will block the dispensing tube.

5.3 CLEANING AND DISINFECTION

Note : Before cleaning the instrument, it is recommended to turn off and disconnect the instrument from the main power supply.

Instrument cleaning

Wipe down all external and internal faces, and plate carrier .

Use of a sponge or a soft, wet cloth with distilled water/detergent solution mixed with disinfection liquid or 70% ethanol.

Note : Avoid chlorinated solvents, hydrocarbon based solvents, and ether.
Avoid abrasive detergent products.

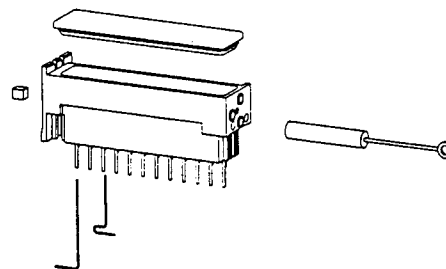
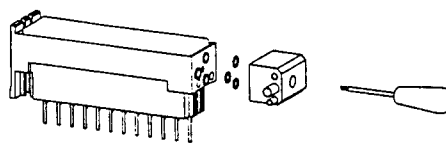
Manifold cleaning

Note : **Manifold is potentially infectious, disinfect first prior to any further handling.**

See Disinfection procedure on next section 5.3.3.

Aspirate needles and chamber

- a) Remove the manifold.
- b) Using a duster or a non scratching utensil, thoroughly clean the outside of the dispense and aspirate needles.
- c) Remove the silicon top seal of the manifold.
- d) Using the big pin supplied in main tenance kit, clean the big aspirating needles.
- e) Rinse out the aspirate chamber with deionized or distilled water. Thoroughly clean the walls of the aspirating cavity.
- f) Look through the aspirate needles to ensure that cleaning is adequate.
- g) Put the silicon top seal back in its place.



Dispense needle and chamber

- a) Push the square rubber cap from the chamber of the manifold using a stem or a screwdriver entered from the opposite end of the cylindrical chamber.
- c) Using one of the small pin supplied in maintenance kit, clean the small dispense needles.
- d) Using the cylindrical brush supplied in maintenance kit, thoroughly clean the small dispense chamber. Thoroughly rinse dispense chamber with deionized water. Ensure that all impurities are removed from the manifold.
- e) Plug dispense chamber of the manifold with the square rubber cap. Place it as close as possible to the first needle however without blocking it. Extra rubber caps are supplied in the maintenance kit.
- f) Remount the manifold, replace the tubing and re-prime.

Fig. 32 Cleaning the Manifold

Manifold	Plate/ Strip	Aspiration Type	Dispense time	Soaking	Cycle	Protected?
8	P = Plate	O = Offset	0.0sec	0 to 9mn 50sec (Plate mode)	0	Yes
or	or	S = Sweep	to	or	to	or
12	S = Strip	C = Centered	9.9 sec	0.0 to 9.9 sec (Strip mode)	9	No

Protocols

1	8	P	S	0.8	0.0	1	No
2	8	P	S	0.0	0.0	0	No
3	8	P	O	0.8	0.0	3	No
4	8	P	O	0.8	0.0	5	No
5	8	S	S	0.8	0.0	1	No
6	8	S	O	0.8	0.0	3	No
7	8	S	O	0.8	0.0	5	No
8	8	P	C	0.0	0.0	0	No
9	8	P	C	0.8	0.0	1	No
10	8	P	C	0.8	0.0	3	No
11	8	S	C	0.8	0.0	3	No
12	8	S	C	0.8	0.0	5	No
13	12	P	S	0.8	0.0	1	No
14	12	P	S	0.0	0.0	0	No
15	12	P	O	0.8	0.0	3	No
16	12	P	O	0.8	0.0	5	No
17	12	S	S	0.8	0.0	1	No
18	12	S	O	0.8	0.0	3	No
19	12	S	O	0.8	0.0	5	No
20	12	P	C	0.0	0.0	0	No
21	12	P	C	0.8	0.0	1	No
22	12	S	C	0.8	0.0	3	No
23	12	S	C	0.8	0.0	5	No
24	12	P	S	0.0	0.0	0	No
25	8	P	S	0.0	0.0	0	No

Comments

1W+A, plate mode, flat bottom, low residual, modifiable
Sweep aspiration, plate mode, flat bottom, low residual, modifiable
3W+A, plate mode, flat bottom, modifiable
5W+A, plate mode, flat bottom, modifiable
1W+A, strip mode, flat bottom, low residual, modifiable
3W+A, strip mode, flat bottom, modifiable
5W+A, strip mode, flat bottom, modifiable
Aspiration, plate mode, curved bottom, modifiable
1W+A, plate mode, curved bottom, modifiable
3W+A, plate mode, curved bottom, modifiable
3W+A, strip mode, curved bottom, modifiable
5W+A, strip mode, curved bottom, modifiable
1W+A, plate mode, flat bottom, low residual, modifiable
Sweep aspiration, plate mode, flat bottom, low residual, modifiable
3W+A, plate mode, flat bottom, modifiable
5W+A, plate mode, flat bottom, modifiable
1W+A, strip mode, flat bottom, low residual, modifiable
3W+A, strip mode, flat bottom, modifiable
5W+A, strip mode, flat bottom, modifiable
Aspiration, plate mode, curved bottom, modifiable
1W+A, plate mode, curved bottom, modifiable
3W+A, strip mode, curved bottom, modifiable
5W+A, strip mode, curved bottom, modifiable
Sweep aspiration, plate mode, flat bottom, low residual, modifiable
Sweep aspiration, plate mode, flat bottom, low residual, modifiable

Disinfection

The disinfection program applies to internal surfaces of tubing, connectors, manifold and manifold needles.

Warning : **Avoid concentrated bleach for disinfecting the washer as a thorough rinsing of the tubes will not remove traces of bleach from tubing and manifold. This can corrupt reagents used in subsequent wash procedures.**

When the washer is used daily, disinfect it at least once a week.

Disinfect the washer prior to any servicing or handling action, shipment or change of site.

- a) Fill the Rinse bottle with disinfection solution approved by your laboratory.
- b) Connect the bottle to the washer and launch a rinse operation. (Fig. 33)

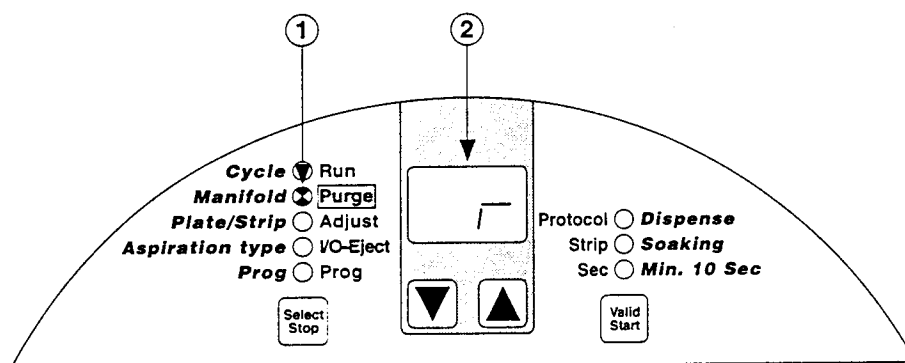


Fig. 33 Rinse tubing and manifold with disinfection solution

- c) After the Rinse operation with disinfection liquid, turn the unit off and wait at least 30 minutes.
- d) In the meantime, disconnect the Rinse bottle from the washer (caution : bottle under pressure) and empty it from the disinfection solution. Wash it thoroughly and fill it with distilled water.
- e) Reconnect Rinse bottle to the washer.
- f) Once the 30 minutes have elapsed, turn the unit on and Rinse 3 or 4 times to eliminate the disinfection solution from the tubing and manifold.

5.4 MEMORY INITIALIZATION

Keep both keys  Select/Stop and  Valid/Start pressed and turn the unit on to reset the memory. All the parameters that might have been modified using the programming mode are reset to their default values.

The memory reset is followed by a beep whereas all the LED's are on for 0.5 sec.



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